

and social finesse needed to successfully navigate our social world. In the words of Jane Jetson, Rosey was “just like one of the family.”

Although the *The Jetsons* correctly predicted cell phones and smartwatches, we still don’t have anything like Rosey. As of this book going to print, even Rosey’s most basic behaviors are still out of reach. It is no secret that the first company to build a robot that can simply *load a dishwasher* will immediately have a bestselling product. All attempts to do this have failed. It isn’t fundamentally a *mechanical* problem; it’s an *intellectual* one—the ability to identify objects in a sink, pick them up appropriately, and load them without breaking anything has proven far more difficult than previously thought.

Of course, even though we do not yet have Rosey, the progress in the field of artificial intelligence (AI) since 1962 has been remarkable. AI can now beat the best humans in the world at numerous games of skill, including chess and Go. AI can recognize tumors in radiology images as well as human radiologists. AI is on the cusp of autonomously driving cars. And as of the last few years, new advancements in large language models are enabling products like ChatGPT, which launched in fall 2022, to compose poetry, translate between languages at will, and even write code. To the chagrin of every high school teacher on planet Earth, ChatGPT can instantly compose a remarkably well written and original essay on almost any topic that an intrepid student might ask of it. ChatGPT can even pass the bar exam, scoring better than 90 percent of lawyers.

Across this long arrow of AI achievements, it has always been hard to tell how close we are to creating human-level intelligence. After the early successes of problem solving algorithms in the 1960s, the AI pioneer Marvin Minsky famously proclaimed that “from three to eight years we will have a machine with the general intelligence of an average human being.” It did not happen. After the successes of expert systems in the 1980s, *BusinessWeek* proclaimed “AI: it’s here.” Progress stalled shortly thereafter. And now with advancements in large language models, many researchers have again proclaimed that the “game is over” because we are “on the verge of achieving human-level AI.” Which is it: Are we finally on the cusp of creating human-like artificial intelligence like Rosey, or are large language models like ChatGPT just the most recent achievement in a long journey that will stretch on for decades to come?

Along this journey, as AI keeps getting smarter, it is becoming harder to measure our progress toward this goal. If an AI system outperforms humans on a task, does it mean that the AI system has captured how humans solve the task? Does a calculator—capable of crunching numbers faster than a human—actually understand math? Does ChatGPT—scoring better on the bar exam than most lawyers—actually understand the law? How can we tell the difference, and in what circumstances, if any, does the difference even matter?

In 2021, over a year before the release of ChatGPT—the chatbot that is now rapidly proliferating throughout every nook and cranny of society—I was using its precursor, a large language model called GPT-3. GPT-3 was trained on large quantities of text (large as in *the entire internet*), and then used this corpus to try to pattern match the most likely response to a prompt. When asked, “What are two reasons that a dog might be in a bad mood?” it responded, “Two reasons a dog might be in a bad mood are if it is hungry or if it is hot.” Something about the new architecture of these systems enabled them to answer questions with what at least seemed like a remarkable degree of intelligence. These models were able to generalize facts they had read about (like the Wikipedia pages about dogs and other pages about causes of bad moods) to new questions they had never seen. In 2021, I was exploring possible applications of these new language models—could they be used to provide new support systems for mental health, or more seamless customer service, or more democratized access to medical information?

The more I interacted with GPT-3, the more mesmerized I became by both its successes and mistakes. In some ways it was brilliant, in other ways it was oddly dumb. Ask GPT-3 to write an essay about eighteenth-century potato farming and its relationship to globalization, and you will get a surprisingly coherent essay. Ask it a commonsense question about what someone might see in a basement, and it answers nonsensically.* Why could GPT-3 correctly answer some questions and not others? What features of human intelligence does it capture, and which is it missing? And why, as AI development continues to accelerate, are some questions that were hard to answer in one year becoming easy in subsequent years? Indeed, as of this book going to print, the new and upgraded version of GPT-3, called GPT-4, released in early 2023, can correctly answer many questions that beguiled GPT-3. And yet still, as we will see in this book, GPT-4 fails to capture

essential features of human intelligence—about something going on in the human brain.

Indeed, the discrepancies between artificial intelligence and human intelligence are nothing short of perplexing. Why is it that AI can crush any human on earth in a game of chess but can't load a dishwasher better than a six-year-old?

We struggle to answer these questions because we don't yet understand the thing we are trying to re-create. All of these questions are, in essence, not questions about AI, but about the nature of human intelligence itself—how it works, why it works the way it does, and as we will soon see, most importantly, how it came to be.

Nature's Hint

When humanity wanted to understand flight, we garnered our first inspiration from birds; when George de Mestral invented Velcro, he got the idea from burdock fruits; when Benjamin Franklin sought to explore electricity, his first sparks of understanding came from lightning. Nature has, throughout the history of human innovation, long been a wondrous guide.

Nature also offers us clues as to how intelligence works—the clearest locus of which is, of course, the human brain. But in this way, AI is unlike these other technological innovations; the brain has proven to be more unwieldy and harder to decipher than either wings or lightning. Scientists have been investigating how the brain works for millennia, and while we have made progress, we do not yet have satisfying answers.

The problem is complexity.

The human brain contains eighty-six billion neurons and over a hundred trillion connections. Each of those connections is so minuscule—less than thirty nanometers wide—that they can barely be seen under even the most powerful microscopes. These connections are bunched together in a tangled mess—within a single cubic millimeter (the width of a single *letter* on a penny), there are over *one billion* connections.

But the sheer number of connections is only one aspect of what makes the brain complex; even if we mapped the wiring of each neuron we would still be far from understanding how the brain works. Unlike the electrical connections in your computer, where wires all communicate using the same